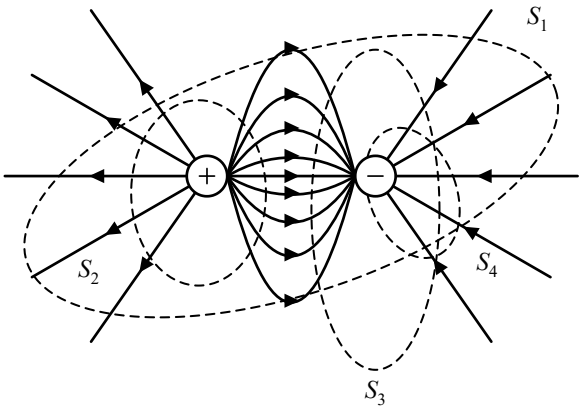
STPM 2016 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

Section A [15 marks]

1. The electric field around an electric dipole is shown in the diagram below.



S_1, S_2, S_3 and S_4 are four Gaussian surfaces. Which surface has **no** net electric flux?

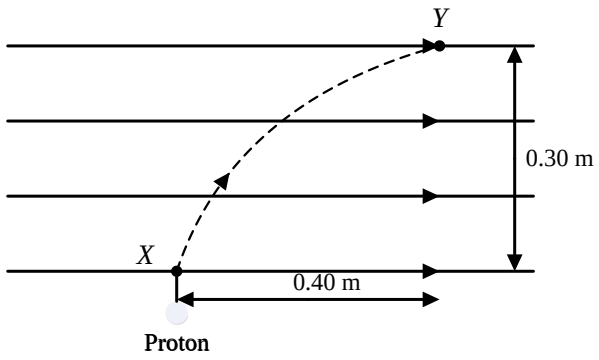
A S_1 B S_2 C S_3 D S_4

Answer: A

S_1 encloses both charges of the dipole: net charge zero, so by Gauss's law the net flux is zero (A).

Explanation: Each of the other surfaces encloses only one charge (or cuts the field lines unequally).

2. A proton enters a uniform electric field at *X* and moves in a parabolic path to *Y* as shown in the diagram below.



If the electric field strength is 500 V m^{-1}, what is the change in kinetic energy of the proton?

A -3.2×10^{-17} J B -2.4×10^{-17} J C $+2.4 \times 10^{-17}$ J D $+3.2 \times 10^{-17}$ J

Answer: D

The displacement along the field from X to Y is 0.40 m; the field does positive work on the proton.

$$\Delta K = qEd = (1.60 \times 10^{-19})(500)(0.40) = +3.2 \times 10^{-17} \text{ J}$$

Explanation: The 0.30 m displacement is perpendicular to the field and does no work.

3. The capacitance of a variable capacitor can be varied from 250 μF to 3750 μF. The capacitance of the capacitor is set at 2000 μF, and is connected to a power source of 12 V. The power source is then removed. What is the maximum potential difference achieved if the capacitance of the capacitor is varied again?

A 6.4 V B 12 V C 23 V D 96 V

Answer: D

$$Q = (2000 \times 10^{-6})(12) = 0.024 \text{ C stays constant; } V_{\max} = \frac{0.024}{250 \times 10^{-6}} = 96 \text{ V}$$

Explanation: The smallest capacitance gives the largest p.d.

4. The drift velocity of free electrons in copper wires is in the order of 10^{-4} m s^{-1} yet the current flow is a few amperes. Which statement explains this effect?

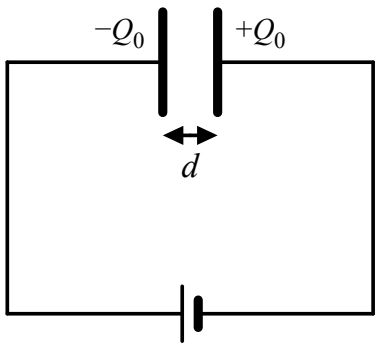
A The current is due to the random velocity but not the drift velocity of the free electrons.
B The free electrons collide frequently with the lattice atoms.
C The density of the number of free electrons in copper wires is very large.
D The current represents the rate of flow of charge across the copper wires.

Answer: C

$I = nAev$: a tiny drift velocity still gives a large current because n is very large (about 10^{29} m^{-3})

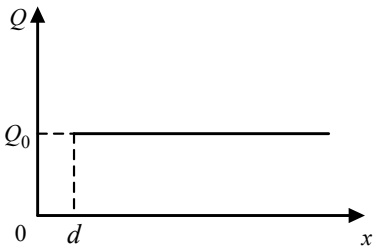
Explanation: The random motion carries no net charge.

5. A parallel-plate capacitor with separation of plates d which is fully charged by a battery to charge Q_0 is shown in the diagram below.

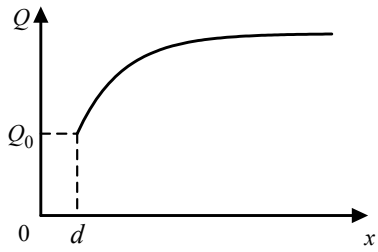


The battery is then disconnected from the capacitor and the distance between the plates x is increased slowly. Which graph shows the variation of the charge Q with x?

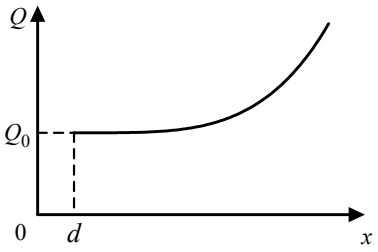
A



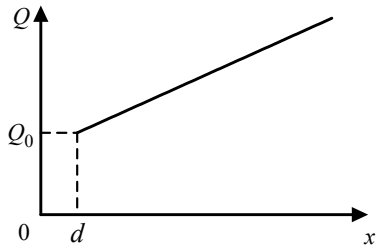
B



C



D



Answer: A

Once disconnected, the capacitor is isolated, so the charge stays at Q_0 whatever the separation (A).

Explanation: C decreases and V increases instead.

6. The element of a kettle is made from a nichrome ribbon of length 5.00 m, width 1.00 mm and thickness 0.0500 mm. When a current of 2.00 A flows in the element, the dissipated power is 500 W. What is the resistivity of the nichrome?

A $2.00 \times 10^{-5}\ \Omega\ \text{m}$ B $2.50 \times 10^{-6}\ \Omega\ \text{m}$ C $1.25 \times 10^{-6}\ \Omega\ \text{m}$ D $1.58 \times 10^{-7}\ \Omega\ \text{m}$

Answer: C

$$R = \frac{P}{I^2} = \frac{500}{2.00^2} = 125\ \Omega;\ A = (1.00 \times 10^{-3})(0.0500 \times 10^{-3}) = 5.00 \times 10^{-8}\ \text{m}^2$$

$$\rho = \frac{RA}{l} = \frac{(125)(5.00 \times 10^{-8})}{5.00} = 1.25 \times 10^{-6}\ \Omega\ \text{m}$$

Explanation: A ribbon has a rectangular cross-section.

7. A battery with an e.m.f. ε and an internal resistance r is connected in series with a resistor of resistance R. The potential difference across the terminal of the battery is

A ε B $\frac{R}{R+r}\varepsilon$ C $\frac{r}{R+r}\varepsilon$ D $\frac{R+2r}{R+r}\varepsilon$

Answer: B

$$I = \frac{\varepsilon}{R+r};\ \text{terminal p.d.} = IR = \frac{R}{R+r}\varepsilon$$

Explanation: The lost volts are Ir.

8. Which statement is true of a voltmeter when it is used to measure the potential difference across an electrical load?

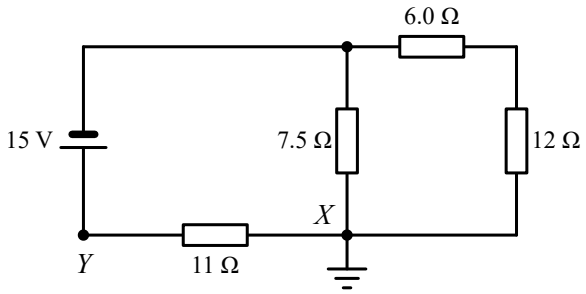
A The current which flows through the voltmeter is negligible.
B The same current flows through the load and the voltmeter.
C The voltmeter has high resistance, thus the effective resistance of the circuit increases.
D The voltmeter has low resistance, thus the effective resistance of the circuit does not change.

Answer: A

A voltmeter has a very high resistance, so the current through it is negligible (A).

Explanation: It is connected in parallel and should not change the circuit.

9. A battery with an e.m.f. of 15 V which is connected to resistors is shown in the circuit below.



If X is earthed, the potential difference at Y is

A −10 V B −5.7 V C 5.7 V D 10 V

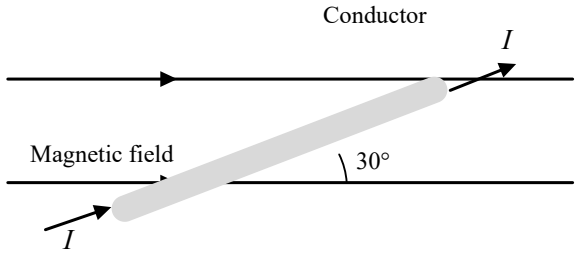
Answer: A

$7.5\ \Omega \parallel (6.0 + 12)\ \Omega = \frac{(7.5)(18)}{25.5} = 5.29\ \Omega$; total $= 5.29 + 11 = 16.3\ \Omega$; $I = \frac{15}{16.3} = 0.921$ A

P.d. across 11 Ω $= (0.921)(11) = 10.1$ V; current flows from X to Y through it, so $V_Y = 0 - 10 = -10$ V

Explanation: Y is joined to the negative terminal of the battery.

10. A conductor of length 0.15 m carrying a constant current I, flowing at an angle of 30° to a magnetic field of 1.8 T is shown in the diagram below.



The cross-sectional area of the conductor is 2.5 cm^2 and the charge carrier density is 3.4×10^{28} m^{-3}. If the magnetic force on the conductor is 4.4 N, what is the drift velocity of free electrons in the conductor?

A 3.6×10^{-6} m s^{-1} B 1.2×10^{-5} m s^{-1} C 1.4×10^{-5} m s^{-1} D 2.4×10^{-5} m s^{-1}

Answer: D

$$F = BIL \sin 30° \Longrightarrow I = \frac{4.4}{(1.8)(0.15)(0.5)} = 32.6 \text{ A}$$

$$v = \frac{I}{nAe} = \frac{32.6}{(3.4 \times 10^{28})(2.5 \times 10^{-4})(1.60 \times 10^{-19})} = 2.4 \times 10^{-5} \text{ m s}^{-1}$$

Explanation: 2.5 $cm^2 = 2.5 \times 10^{-4}$ m^2.

11. A solenoid of length 2.00 m has 1500 turns. The cross-sectional area of the solenoid is 5.0×10^{-4} m^2. What is the self-inductance of the solenoid?

A 4.71×10^{-7} H B 1.88×10^{-6} H C 7.07×10^{-4} H D 2.83×10^{-3} H

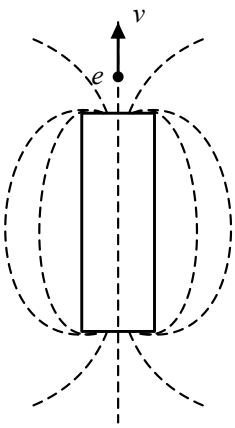
Answer: C

$$L = \frac{\mu_0 N^2 A}{l} = \frac{(4\pi \times 10^{-7})(1500)^2(5.0 \times 10^{-4})}{2.00} = 7.07 \times 10^{-4} \text{ H}$$

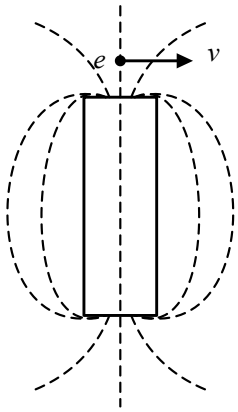
Explanation: Square the number of turns.

12. An electron moves with a speed v near a bar magnet. Which diagram shows the electron e experiencing the maximum magnetic force?

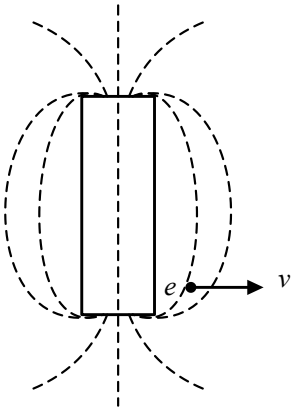
A



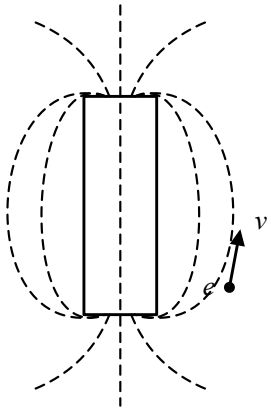
B



C



D



Answer: B

The force $evB \sin \theta$ is largest where **B** is strongest (near a pole) and the velocity is perpendicular to the field: diagram B.

Explanation: Moving along a field line (A, D) gives no force; C is in a weaker field.

13. A sinusoidal a.c. voltage is applied across a resistor. Which statement is true about the current and the voltage of the resistor?

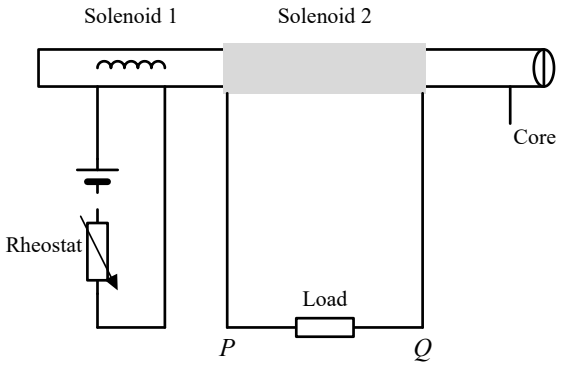
A The current is maximum when the voltage is zero.
B The current is zero when the voltage is maximum.
C The current increases when the voltage increases.
D The current increases when the voltage decreases.

Answer: C

In a resistor the current and voltage are in phase: the current increases when the voltage increases (C).

Explanation: Both are maximum and zero at the same instants.

14. Two solenoids which have a common core are shown in the diagram below.



The rheostat is adjusted to reduce its resistance gradually. Which statement explains what will happen to the components in the diagram?

A The magnetic field in the core decreases.
B The induced current in the load flows from P to Q.
C No e.m.f. is induced in solenoid 2 if the adjustment is done at a constant rate.
D Induced e.m.f. in solenoid 2 increases if the adjustment is done at a slower rate.

Answer: B

Reducing the rheostat resistance increases the current and the flux in the core, so an e.m.f. is induced in solenoid 2.

By Lenz's law the induced current opposes the increasing flux; for the windings shown it flows through the load from P to Q (B).

Explanation: A is wrong (the field increases); an e.m.f. is induced whenever the current changes; a slower change gives a smaller e.m.f.

15. Sinusoidal voltage with a peak voltage V_0 is connected across a resistor of resistance R. The average power which is dissipated as heat is

A 0 B $\frac{V_0^2}{2R}$ C $\frac{V_0^2}{R}$ D $\frac{2V_0^2}{R}$

Answer: B

$$P = \frac{V_{\text{rms}}^2}{R} = \frac{(V_0/\sqrt{2})^2}{R} = \frac{V_0^2}{2R}$$

Explanation: The mean of $\sin^2$ over a cycle is ½.

Section B [15 marks]

16. (a) A conducting wire with a cross-sectional area A has a number of electrons per unit volume n. The electrons have a drift velocity v_d and a charge q. Derive an expression for the current which flows through the wire in terms of A, n, v_d and q. [3]

Answer

In a length l of wire, the number of electrons = nAl, carrying charge $Q = nAlq$ — 1

They all pass a cross-section in time $t = \frac{l}{v_d}$ — 1

$I = \frac{Q}{t} = nAv_dq$ — 1

3

Explanation: Current is the rate of flow of charge.

(b) A copper wire with a diameter of 4.25 mm carries a constant current of 8.00 A. The number of electrons per unit volume is 8.50×10^{28} m^{-3}.

(i) Determine the drift velocity of the electrons. [2]

Answer

$A = \pi(2.125 \times 10^{-3})^2 = 1.42 \times 10^{-5}$ m^2; $v_d = \frac{8.00}{(8.50 \times 10^{28})(1.42 \times 10^{-5})(1.60 \times 10^{-19})}$ — 1

$v_d = 4.15 \times 10^{-5}$ m s^{-1} — 1

2

Explanation: Use the radius, half of 4.25 mm.

(ii) Determine the current density in the copper wire. [2]

Answer

$J = \frac{I}{A} = \frac{8.00}{1.42 \times 10^{-5}}$ — 1

$J = 5.64 \times 10^5$ A m^{-2} — 1

2

Explanation: Current per unit cross-sectional area.

(iii) The copper wire is stretched while the current is kept constant. What happens to the magnitude of the current density? [1]

Answer

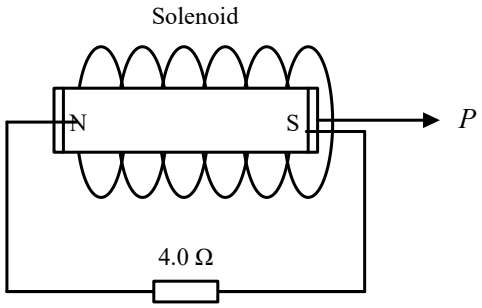
It increases, because stretching reduces the cross-sectional area while I is unchanged — 1

1

Total for question **8**

Explanation: $J = I/A$.

17. A solenoid of negligible resistance is connected to a resistor of 4.0 Ω. A bar magnet which is placed inside the solenoid is shown in the diagram below.



(a) Explain why current is induced in the circuit when the bar magnet is pulled out in the direction of P. [2]

Answer

As the magnet is pulled out, the magnetic flux linking the solenoid changes (decreases) — 1

An e.m.f. is induced (Faraday's law), and since the circuit is closed, a current flows — 1

2

Explanation: No relative motion, no induced current.

(b) Using Lenz's law, explain how the direction of the induced current in the resistor can be determined. [2]

Answer

By Lenz's law the induced current opposes the change: it tries to attract the departing magnet (S pole leaving), so the end of the solenoid near P becomes an N pole — 1

The right-hand grip rule then gives the current direction in the solenoid and hence through the resistor — 1

2

Explanation: The induced current opposes the motion of the magnet.

(c) If the initial magnetic flux linkage which passes through the solenoid is 4.00 Wb and the bar magnet is pulled out completely from the solenoid in 2.00 s, determine the average induced current which flows through the resistor. [3]

Answer

$\varepsilon = \frac{\Delta(N\Phi)}{\Delta t} = \frac{4.00}{2.00} = 2.00$ V — 1

$I = \frac{\varepsilon}{R} = \frac{2.00}{4.0}$ — 1

$I = 0.50$ A — 1

3

Total for question **7**

Explanation: The flux linkage falls from 4.00 Wb to zero.

Section C [30 marks]

18. (a) Define *electric field strength* and *electric potential*. [2]

Answer

Electric field strength: the force per unit positive charge at that point ($E = F/q$) — 1

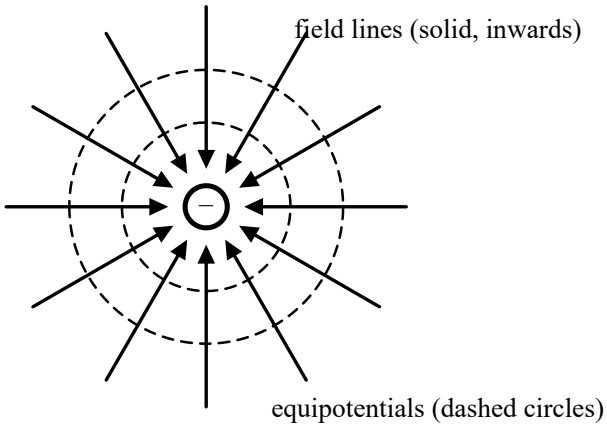
Electric potential: the work done per unit positive charge in bringing a positive test charge from infinity to that point — 1

2

Explanation: E is a vector; V is a scalar.

(b) Sketch the electric field and the equipotential surfaces of a negative point charge. [4]

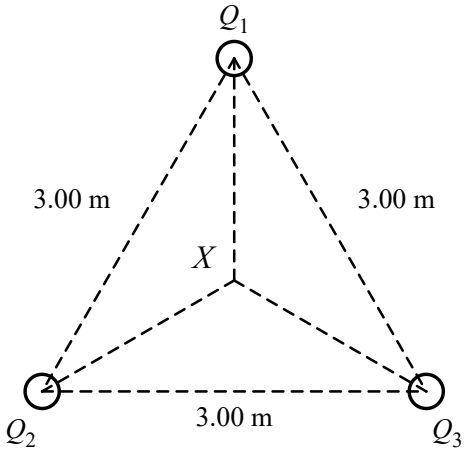
Answer



Radial field lines	1
Arrows pointing towards the negative charge	1
Equipotentials as concentric circles (spheres) centred on the charge	1
Equipotentials perpendicular to the field lines and further apart at larger distances	1
	4

Explanation: The potential is negative and increases (towards zero) with distance.

(c) Point charges Q_1, Q_2 and Q_3 are located at the vertices of an equilateral triangle as shown in the diagram below.



Given that Q_1 is +45.0 μC, Q_2 is −12.0 μC and Q_3 is +18.0 μC, determine the magnitude and direction of the

(i) electrostatic force on Q_2, [5]

Answer

$F_{12} = \frac{(9.0 \times 10^9)(45.0 \times 10^{-6})(12.0 \times 10^{-6})}{(3.00)^2} = 0.54$ N, attraction towards Q_1 1

$F_{32} = \frac{(9.0 \times 10^9)(18.0 \times 10^{-6})(12.0 \times 10^{-6})}{(3.00)^2} = 0.216$ N, attraction towards Q_3 1

Components (along Q_2Q_3, perpendicular): $F_x = 0.54 \cos 60° + 0.216 = 0.486$ N, $F_y = 0.54 \sin 60° = 0.468$ N 1

$F = \sqrt{(0.486^2 + 0.468^2)} = 0.67$ N 1

At $\tan^{-1}(0.468/0.486) = 44°$ above Q_2Q_3, between the directions of Q_3 and Q_1 1

5

Explanation: Both forces on the negative Q_2 are attractions.

(ii) electric potential at the centre point X of the triangle. [4]

Answer

Distance from each vertex to the centre: $r = \frac{3.00}{\sqrt{3}} = 1.73$ m 1

$V = \frac{k(Q_1 + Q_2 + Q_3)}{r} = \frac{(9.0 \times 10^9)(45.0 - 12.0 + 18.0) \times 10^{-6}}{1.73}$ 1

$V = 2.65 \times 10^5$ V 1

Potential is a scalar, so it has no direction 1

4

Total for question **15**

Explanation: All three charges are the same distance from X.

19. (a) Define *capacitance* of a capacitor. [1]

Answer

The ratio of the charge on either plate to the p.d. between the plates, $C = \frac{Q}{V}$ 1

1

Explanation: Unit: farad.

(b) Derive an expression of an effective capacitance for two capacitors which are connected in series. [2]

Answer

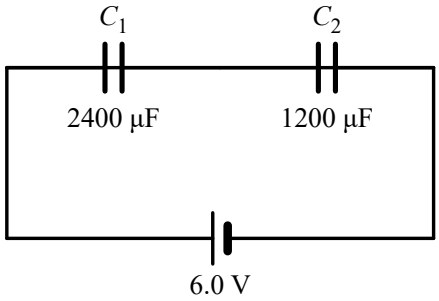
In series each capacitor carries the same charge Q, and $V = V_1 + V_2$ 1

$\frac{Q}{C} = \frac{Q}{C_1} + \frac{Q}{C_2} \Rightarrow \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$ 1

2

Explanation: The effective capacitance is less than either.

(c) Capacitors C_1 and C_2 which are connected in series with a battery are shown in the diagram below.



(i) Determine the charge stored in each capacitor. [3]

Answer

$C = \frac{(2400)(1200)}{2400 + 1200} = 800\ \mu F$ 1

$Q = CV = (800 \times 10^{-6})(6.0)$ 1

$Q = 4.8 \times 10^{-3}$ C on each capacitor 1

3

Explanation: Series capacitors carry equal charges.

(ii) Determine the total energy stored in the capacitors. [2]

Answer

$U = \frac{1}{2}CV^2 = \frac{1}{2}(800 \times 10^{-6})(6.0)^2$ 1

$U = 1.44 \times 10^{-2}$ J 1

2

Explanation: Or the sum of $\frac{1}{2}Q^2/C$ for each.

(iii) Calculate the energy supplied by the battery. [2]

Answer

$W = QV = (4.8 \times 10^{-3})(6.0)$ 1

$W = 2.88 \times 10^{-2}$ J 1

2

Explanation: The battery moves all the charge through 6.0 V.

(iv) Comment on your answers in (c)(ii) and (c)(iii). [2]

Answer

Only half of the energy supplied by the battery is stored in the capacitors 1

The other half is dissipated as heat in the resistance of the connecting wires during charging 1

2

Explanation: This is independent of the resistance.

(v) Capacitor C_1 is removed and allowed to discharge through a resistor of 250 Ω. Determine the time taken for the charge to decrease by 60.0%. [3]

Answer

$\tau = RC_1 = (250)(2400 \times 10^{-6}) = 0.60$ s 1

Charge decreases by 60%: $Q = 0.40Q_0 = Q_0 e^{-t/\tau}$ 1

$t = 0.60 \ln \frac{1}{0.40} = 0.55$ s 1

3

Total for question **15**

Explanation: "Decrease by 60%" means 40% remains.

20. (a) Define *impedance*. [1]

Answer

The ratio of the r.m.s. (or peak) voltage to the r.m.s. (or peak) current in an a.c. circuit, $Z = V/I$ 1

1

Explanation: It combines resistance and reactance.

(b) An alternating voltage is given by $V = 340 \sin 314t$, where V is measured in volts and t in seconds. This voltage is supplied across a pure capacitor of 220 μF.

(i) Determine the r.m.s. voltage. [3]

Answer

$V_0 = 340$ V — 1

$V_{\text{rms}} = \frac{V_0}{\sqrt{2}} = \frac{340}{\sqrt{2}}$ — 1

$V_{\text{rms}} = 240$ V — 1

3

Explanation: For a sinusoidal voltage.

(ii) Derive an expression for the current I which flows through this capacitor. [3]

Answer

$Q = CV \Longrightarrow I = \frac{\mathrm{d}Q}{\mathrm{d}t} = C\frac{\mathrm{d}V}{\mathrm{d}t}$ — 1

$I = (220 \times 10^{-6})(340)(314) \cos 314t$ — 1

$I = 23.5 \cos 314t$ A — 1

3

Explanation: The current leads the voltage by 90°.

(iii) Determine the reactance for the capacitor. [2]

Answer

$X_C = \frac{1}{\omega C} = \frac{1}{(314)(220 \times 10^{-6})}$ — 1

$X_C = 14.5\ \Omega$ — 1

2

Explanation: Check: $V_0/I_0 = 340/23.5 = 14.5\ \Omega$.

(iv) A resistor is connected in series to the capacitor so that the current is reduced. Determine the value of resistance of the resistor which must be connected in order for the r.m.s. current I to be reduced to 2.4 A. [3]

Answer

$Z = \frac{V_{\text{rms}}}{I_{\text{rms}}} = \frac{240}{2.4} = 100\ \Omega$ 1

$R = \sqrt{(Z^2 - X_C^2)} = \sqrt{(100^2 - 14.5^2)}$ 1

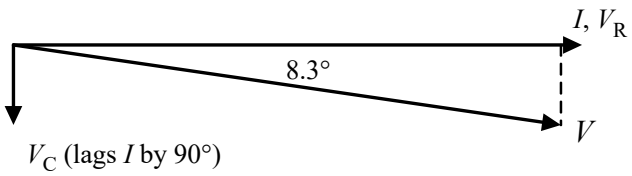
$R = 98.9\ \Omega$ 1

3

Explanation: $Z^2 = R^2 + X_C^2$.

(v) Determine the phase angle between the current I and the voltage V. Sketch a phasor diagram for this R-C circuit. [3]

Answer



$\tan \varphi = \frac{X_C}{R} = \frac{14.5}{98.9} \Rightarrow \varphi = 8.3°$ 1

The current leads the voltage 1

Phasor diagram: V_R along I, V_C 90° behind I, V their resultant 1

3

Total for question **15**

Explanation: The circuit is mainly resistive, so the phase angle is small.